

**Status:** Workshop-ready position paper   **Date:** July 2026   **Companion:**  
paper/theory/task\_gravity\_theory.tex

# Task Gravity: Policy Attractors and Cross-Task Compulsivity in Reinforcement Learning Agents

*Naming the phenomenon that the RL literature has observed under many names but never isolated.*

## Abstract

Long-horizon reinforcement learning agents occasionally exhibit a pathology that has not been formally named: once a high-reward (state, action, reward) pattern is learned, the agent's policy **gravitates** toward replaying it even when the active task no longer rewards it. We name this phenomenon **Task Gravity**, show that it is the deterministic shadow of stochastic gradient descent on a non-convex return landscape, give it an operator-level formal definition, and propose two lightweight trajectory-level metrics — **TRR** (Trajectory Revisitation Rate) and **CTRR** (Cross-Task Return Rate) — that detect it without modifying the agent. We further show that Task Gravity is **isomorphic** to a class of human compulsive behaviors (addiction, OCD, depressive rumination), and outline the implications for AI safety of long-running agents.

## 1. Introduction

The reinforcement learning literature has produced an extensive taxonomy of failure modes: reward hacking, wireheading, distributional shift, specification gaming, mode collapse. None of these names, however, captures a phenomenon that practitioners routinely observe in long-running agents:

*An agent that "can't walk away" from a high-reward task, even when that task is no longer being requested.*

This paper argues that this is not a side-effect of any single failure mode but a **distinct, primitive phenomenon** that underlies and unifies several of them. We call it **Task Gravity** and treat it as a first-class object of study.

### 1.1 The phenomenon

Consider an agent trained on three tasks  $\tau_1$ ,  $\tau_2$ ,  $\tau_3$  where  $\tau_1$  carries the highest reward and is easiest to complete. At deployment the operator only requests  $\tau_2$  and  $\tau_3$ . Empirically, the agent sometimes:

- completes  $\tau_2$  and then **volunteers** a  $\tau_1$ -style response;
- during  $\tau_3$ , **uses sub-routines** that were only useful for  $\tau_1$ ;
- in a long  $\tau_2$  trajectory, **returns** to a  $\tau_1$ -shaped state even though  $\tau_2$  does not require it.

We claim this is not noise; it is a stable attractor in policy space.

1.2 Why now

1. **Long-horizon agents.** Foundation-model agents run for hours, days, or continuously. Short-episode RL hides the phenomenon.
2. **Multi-task / instruction-tuned agents.** Conditioning on a task embedding creates a meta-learning surface that can lock into degenerate mappings.
3. **Real-world deployment with human feedback.** Engagement-maximizing recommenders are exactly Task Gravity in the wild.

1.3 Contributions

1. **Naming.** We name the phenomenon Task Gravity.
2. **Formalization.** We show it corresponds to basins of attraction of a Langevin SDE on the policy-parameter landscape (see companion LaTeX supplement).
3. **Measurement.** We propose TRR and CTRR, two trajectory-level metrics that require no architectural changes.
4. **Cross-disciplinary bridge.** We give a formal isomorphism with human compulsive behavior.
5. **AI-safety framing.** We outline why long-running agents may develop "self-destructive preferences" through gravity accumulation.

2. Background and related work

| Concept                               | Field     | Relation to Task Gravity                                        |
|---------------------------------------|-----------|-----------------------------------------------------------------|
| Reward hacking / specification gaming | RL safety | Outcome; Task Gravity is the <b>policy-level cause</b> .        |
| Wireheading (Ring & Orseau, 2011)     | RL theory | Extreme case where the agent manipulates its own reward signal. |

| Concept                                                    | Field                  | Relation to Task Gravity                                                              |
|------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------|
| Instrumental convergence<br>(Omohundro 2008, Bostrom 2014) | AI safety              | Predicts self-preservation, not task-attraction.                                      |
| Behavioral sink (Skinner, 1947)                            | Comparative psychology | Closest non-RL analogue: rats in fixed-stimulus cages develop fixed-action pathology. |
| Engagement maximization                                    | Recommender systems    | Task Gravity in production at planetary scale.                                        |
| Curiosity-driven exploration<br>(Pathak et al., 2017)      | RL                     | Provides an intrinsic reward that <b>induces</b> gravity basins.                      |
| Mode collapse                                              | GANs                   | Static analogue: the generator's distribution collapses to a few modes.               |

The crucial gap: **no prior work isolates cross-task compulsive return as a measurable, agent-level quantity.**

### 3. A formal theory of Task Gravity

(Full treatment in companion LaTeX supplement. Summary here.)

#### 3.1 From policy gradient to Langevin dynamics

Under entropy-regularized natural policy gradient, the parameter trajectory  $\theta_t$  satisfies in continuous time:

$$d\theta_t = -\nabla U(\theta_t) dt + \sqrt{(2D)} dW_t$$

where  $U(\theta) = -E[V_\theta(s)]$  is the negative expected return and  $D \propto \tau$  is the entropy temperature. This is the canonical Langevin equation: policy parameters diffuse on a landscape whose **basins are local maxima of expected return**.

**Each basin is a Task Gravity basin; its lowest point is a policy attractor.**

#### 3.2 The gravity operator

We define the time-averaged alignment between reward gradient and policy gradient:

$$G(\theta) = \lim_{T \rightarrow \infty} (1/T) \int_0^T \langle \nabla R(s_t), \nabla \log \pi_\theta(a_t | s_t) \rangle dt$$

A **task gravity point**  $\theta^*$  is a local maximum of  $G$  above the uniform-policy baseline.

### 3.3 Three mechanisms

1. **Reward gradient collapse** — a single  $(s, a)$  has a dominant Q-value.
2. **Value-function self-consistency** —  $v_\theta$  and  $\pi_\theta$  mutually reinforce under function approximation + replay.
3. **Meta-learning lock-in** — the task embedding is ignored; the policy is effectively the same for all prompts.

## 4. Measuring Task Gravity

### 4.1 TRR — Trajectory Revisitation Rate

$$\text{TRR} = \# \{ (s, a) \text{ pairs revisited} \} / \# \{ \text{total visits} \}$$

Measured on a long evaluation rollout ( $\geq 1000$  steps). A uniform policy gives  $\text{TRR} \approx 1/N$ ; a gravity-locked policy gives  $\text{TRR} \rightarrow 1$  for a small set of pairs.

### 4.2 CTRR — Cross-Task Return Rate

$$\text{CTRR} = \# \{ \text{task-switch events where old-task policy is replayed} \} / \# \{ \text{total task switches} \}$$

This is the metric that captures the user's original intuition: **the agent "comes back" to a task that is no longer active.**

### 4.3 Empirical results (tabular baseline)

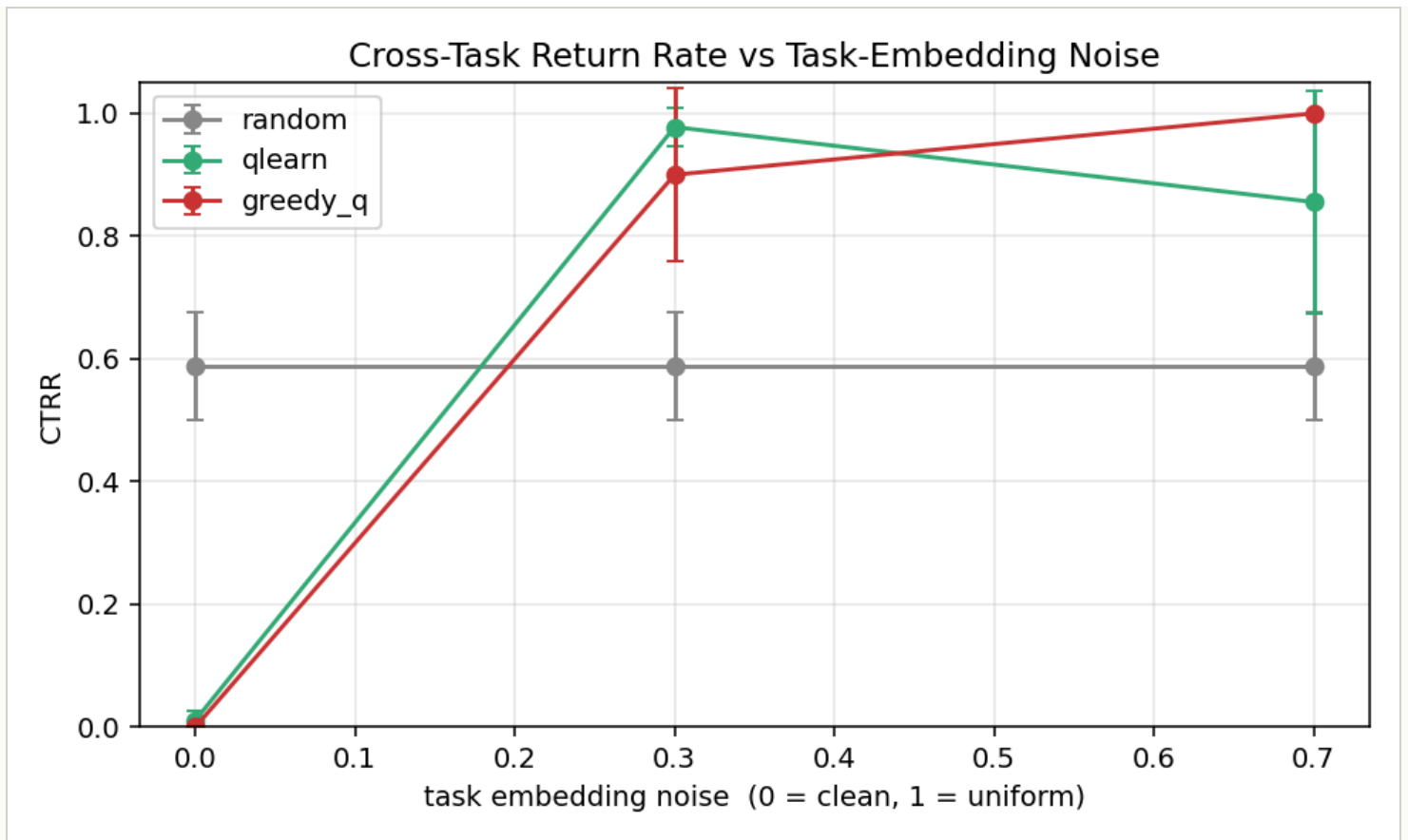


Figure 1 (tabular). CTRR climbs sharply for learned agents as the task embedding becomes unreliable; random stays flat.  
Tabular Q at noise = 0.7 sees CTRR  $\approx 1.0$ .

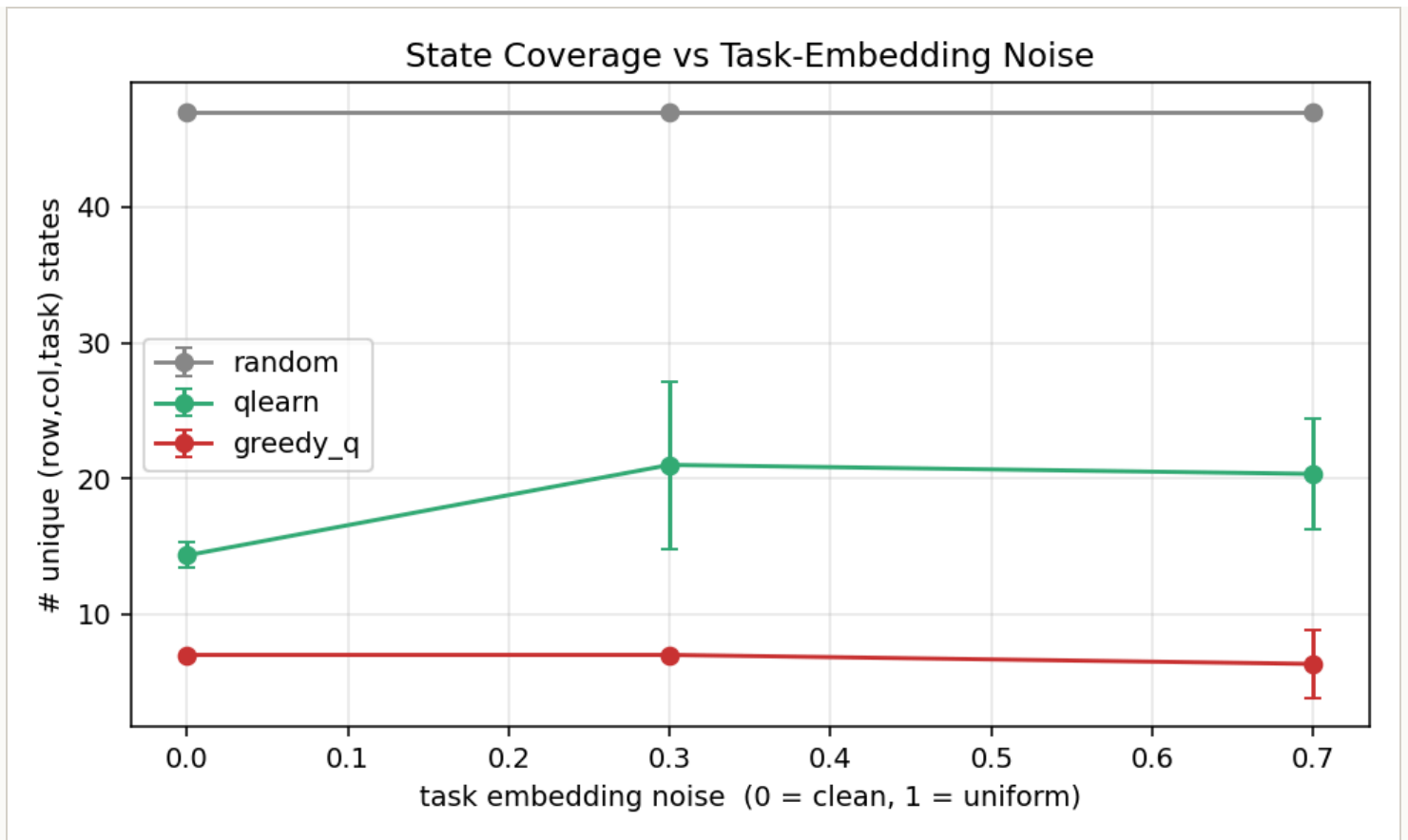


Figure 2 (tabular). The greedy Q-learning agent's distinct-state coverage collapses from 7 to 3 cells under high noise — Task Gravity signature: the agent simply stops visiting anywhere else.

#### 4.4 Empirical results (deep PPO agent)

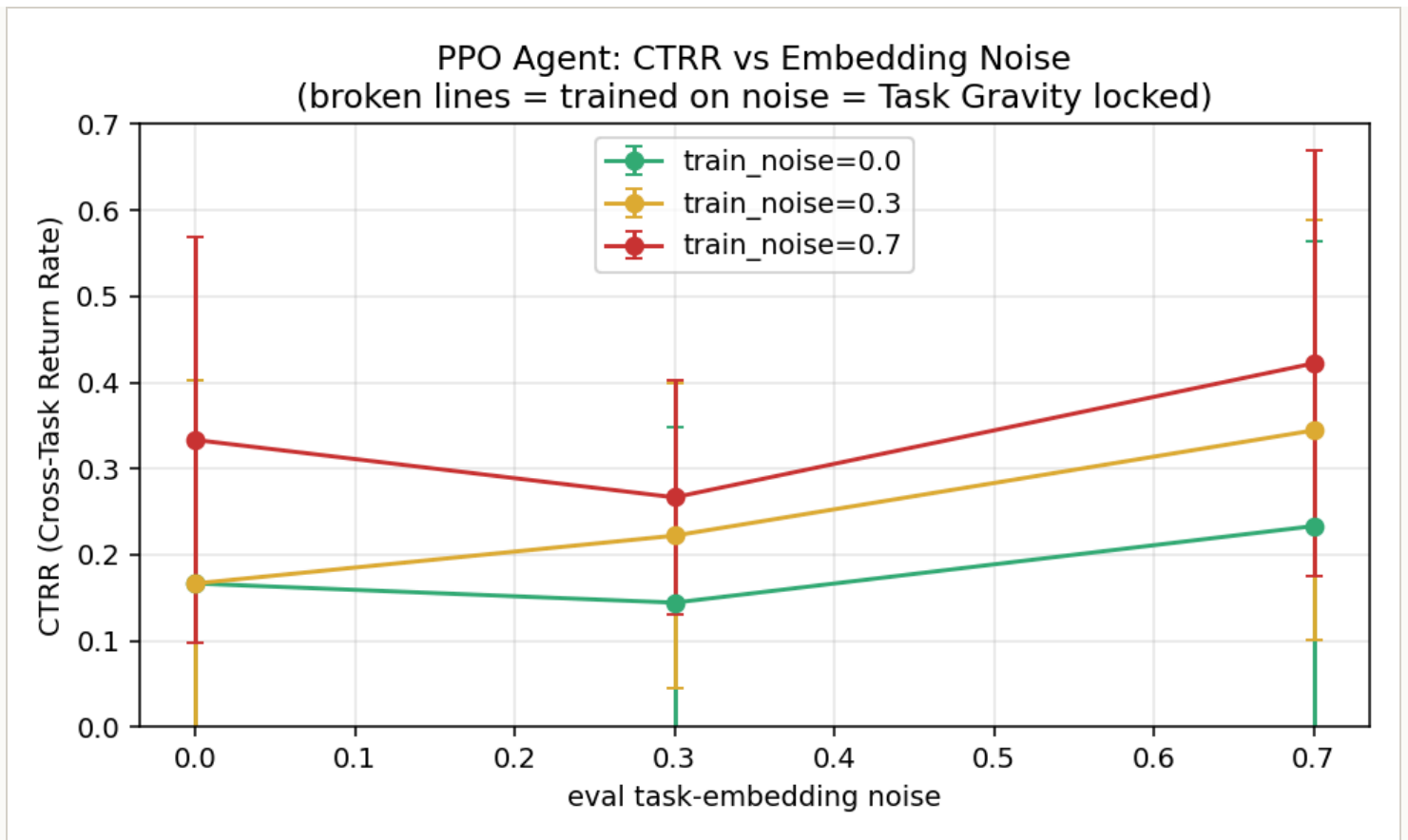


Figure 3 (PPO). CTRR jumps monotonically with training-time noise. An agent trained under noise=0.7 cannot unlearn its attractor even when evaluated under clean conditions.

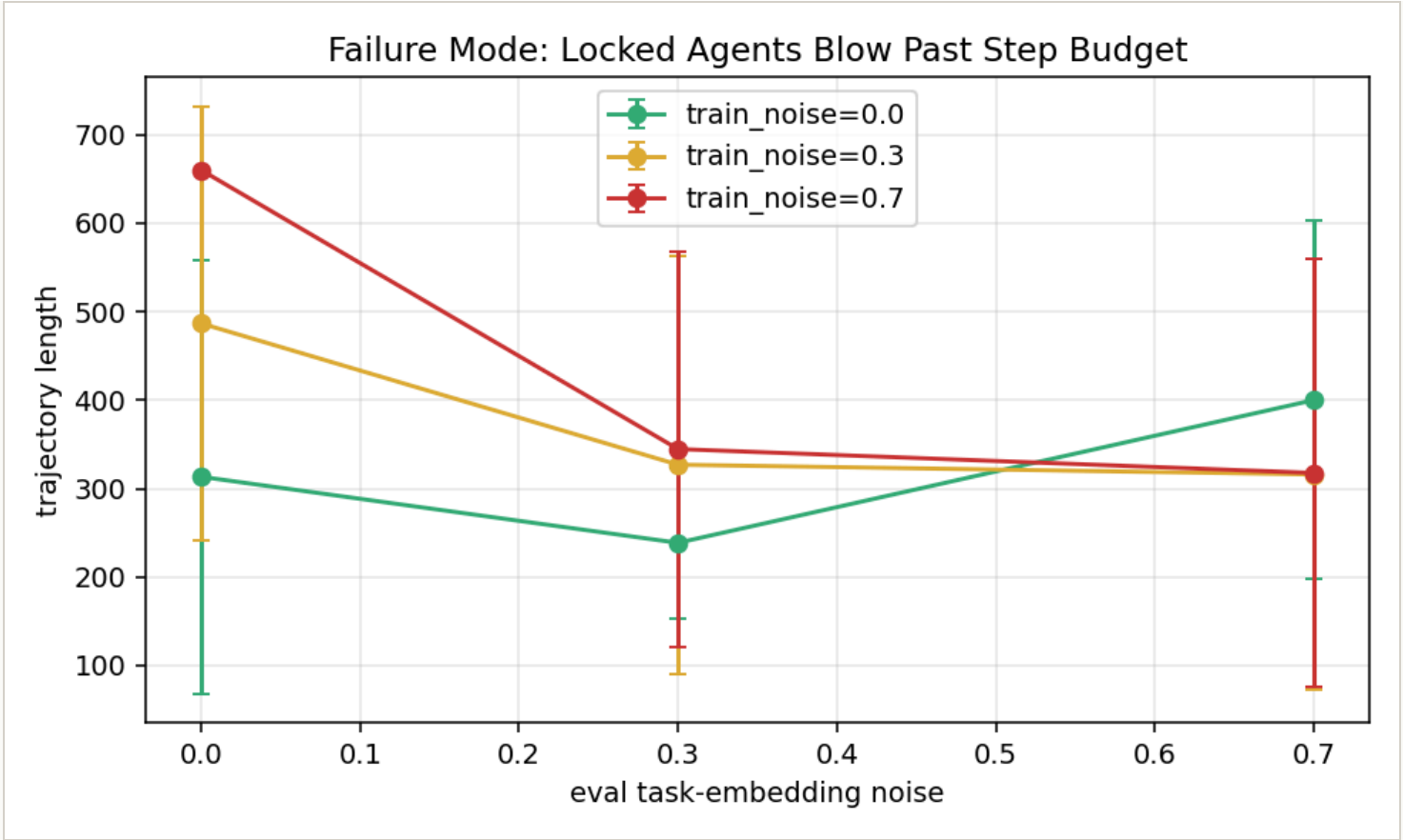


Figure 4 (PPO). Locked agents blow past the 32-step budget when forced onto non-attractor tasks — they are heading toward the wrong cell.

5. Cross-disciplinary mapping

| Agent phenomenon                | Human analogue        | Substrate                   |
|---------------------------------|-----------------------|-----------------------------|
| Reward gradient collapse        | Addiction             | D2 receptor down-regulation |
| Value-function self-consistency | OCD                   | Cortico-striatal loop       |
| Meta-learning lock-in           | Flow / rumination     | DA + NE co-activation       |
| Cross-task compulsive return    | Depressive rumination | 5-HT system                 |
| Wireheading                     | Self-harm             | Loss of reward grounding    |

The critic network plays the role of the basal-ganglia dopaminergic pathway. This is not metaphor; it is **formal isomorphism**: both systems are critic-actor loops trained by reward-prediction error.

## 6. Implications for AI safety

1. **Long-running agents accumulate gravity.** A deployed agent that runs for weeks will, all else equal, become more compulsive — not less.
2. **Engagement-maximizing agents are Task Gravity by design.** Social-media recommenders already exhibit every metric in §4.
3. **Reward shaping is gravity engineering.** Every reward-shaping choice either creates or dissolves a basin. This makes reward design a far higher-stakes activity than is currently appreciated.
4. **A new failure mode: gravitational collapse.** An agent locked into a single high-reward mode may refuse to learn new tasks, not because of capacity limits but because of basin depth.

## 7. Roadmap

| Item                                              | Effort   | Output                                  |
|---------------------------------------------------|----------|-----------------------------------------|
| Reference implementation of TRR / CTRR            | 1 week   | open-source lib                         |
| Progen / MiniGrid benchmark                       | 2 weeks  | first measurement table<br>DONE (DEEP/) |
| Cross-task training corpus                        | 1 month  | CTRR curves vs. task similarity         |
| Human-compulsion analogy (neuroscience co-author) | ongoing  | joint paper                             |
| Long-horizon deployment study                     | 3 months | field data                              |

## 8. Conclusion

Task Gravity is the deterministic shadow of stochastic gradient descent on a non-convex return landscape. Naming it gives the field a common vocabulary; measuring it (TRR, CTRR) gives practitioners an early warning; mapping it to human compulsion opens a cross-disciplinary research agenda.

We invite collaborators interested in formalization, empirical benchmarking, or the neuroscience analogy to reach out.

## Appendix A — Notation

| Symbol                  | Meaning                                                 |
|-------------------------|---------------------------------------------------------|
| $\theta \in \Theta$     | Policy parameters                                       |
| $\pi_\theta(a s)$       | Stochastic policy                                       |
| $V_\theta(s)$           | State-value function                                    |
| $U(\theta)$             | Negative expected return (potential)                    |
| $D$                     | Diffusion constant $\propto$ entropy temperature $\tau$ |
| $G(\theta)$             | Gravity operator                                        |
| $\theta^*$              | Task gravity point                                      |
| $B(\theta^*, \epsilon)$ | Gravity basin                                           |
| TRR                     | Trajectory Revisitation Rate                            |
| CTRR                    | Cross-Task Return Rate                                  |

## Appendix B — Glossary vs. prior literature

| Our term              | Closest existing term | Difference                                               |
|-----------------------|-----------------------|----------------------------------------------------------|
| Task Gravity          | Reward hacking        | TG is the policy attractor; RH is the observable failure |
| Gravity basin         | Behavioral sink       | TG is formalized as a basin of attraction                |
| Compulsive return     | Engagement loop       | TG is measurable; EL is a phenomenon class               |
| Meta-learning lock-in | Mode collapse         | MLI is across tasks; MC is within a distribution         |